

Exercise 2: E-commerce Platform Search Function

CODE :

```
using System;

class Product{

    public int ProductId { get; set; }

    public string ProductName { get; set; }

    public string Category { get; set; }

    public Product(int id, string name, string category){

        ProductId = id;

        ProductName = name;

        Category = category;}}

class SearchDemo{

    static Product LinearSearch(Product[] products, int productId){

        foreach (Product p in products){

            if (p.ProductId == productId){

                return p;}}

        return null;}

    static Product BinarySearch(Product[] products, int productId){

        int left = 0;

        int right = products.Length - 1;

        while (left <= right){

            int mid = (left + right) / 2;

            if (products[mid].ProductId == productId){

                return products[mid];}

            else if (products[mid].ProductId < productId){

                left = mid + 1;}

            else{

                right = mid - 1;}}
```

```
        return null; }

static void Main(){

    Product[] products ={

        new Product(105, "Laptop", "Electronics"),

        new Product(101, "Shoes", "Fashion"),

        new Product(110, "Phone", "Electronics"),

        new Product(103, "Watch", "Accessories"),

        new Product(108, "Bag", "Fashion")};

    Product[] sortedProducts = {

        new Product(101, "Shoes", "Fashion"),

        new Product(103, "Watch", "Accessories"),

        new Product(105, "Laptop", "Electronics"),

        new Product(108, "Bag", "Fashion"),

        new Product(110, "Phone", "Electronics")

    };

    int searchId = 108;

    Console.WriteLine("Linear Search:");

    Product result1 = LinearSearch(products, searchId);

    if (result1 != null)

        Console.WriteLine($"Found: {result1.ProductName}");

    else

        Console.WriteLine("Product not found");

    Console.WriteLine();

    Console.WriteLine("Binary Search:");

    Product result2 = BinarySearch(sortedProducts, searchId);

    if (result2 != null)

        Console.WriteLine($"Found: {result2.ProductName}");

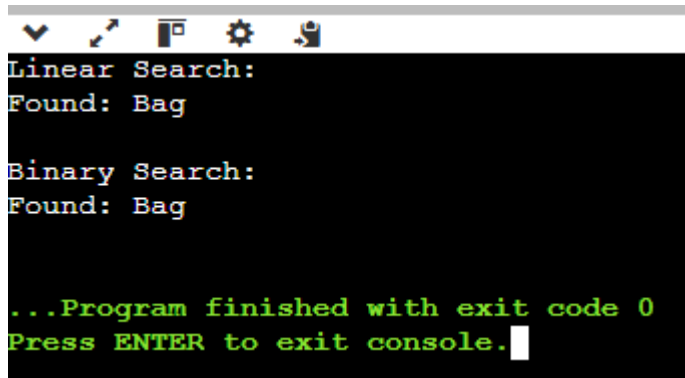
    else
```

```

        Console.WriteLine("Product not found");
    }
}

```

OUTPUT :



```

Linear Search:
Found: Bag

Binary Search:
Found: Bag

...Program finished with exit code 0
Press ENTER to exit console.

```

Exercise 7: Financial Forecasting

CODE :

```

using System;

class FinancialForecast{

    static double PredictFutureValue(double currentValue,double growthRate,int years){

        if (years == 0){

            return currentValue;}

        return PredictFutureValue(currentValue,growthRate,years - 1) * (1 + growthRate);

    }

    static void Main()

    {

        double currentValue = 10000;

        double growthRate = 0.10; // 10%

        int years = 5;

        double futureValue =

            PredictFutureValue(currentValue,

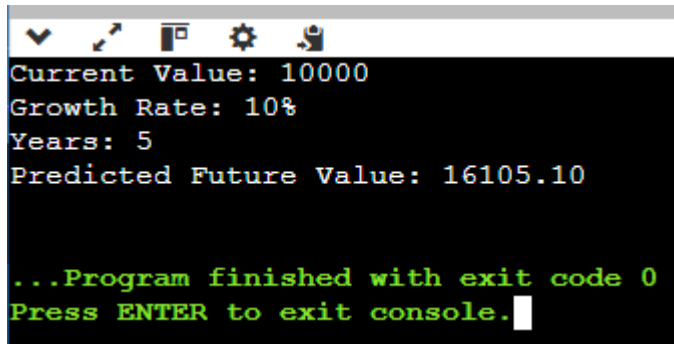
                                growthRate,

                                years);
    }
}

```

```
Console.WriteLine("Current Value: " + currentValue);  
  
Console.WriteLine("Growth Rate: " + (growthRate * 100) + "%");  
  
Console.WriteLine("Years: " + years);  
  
Console.WriteLine("Predicted Future Value: " +  
    futureValue.ToString("F2"));  
  
}  
}
```

OUTPUT :



```
Current Value: 10000  
Growth Rate: 10%  
Years: 5  
Predicted Future Value: 16105.10  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```